HW1-ICA :: Part A

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This notebook:

- Loads `cat.png` using OpenCV
- Converts the image from BGR to RGB
- Plots the RGB image using Matplotlib

All paths are relative to the notebook location (project root).

In [26]:
```
from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt

try:
    import cv2
except ImportError as e:
    raise ImportError(
        "OpenCV is not installed."
    ) from e

print("cv2 version:", cv2.__version__)
```

```
cv2 version: 4.10.0
```

In [27]:
```
# Here we load image
# OpenCV reads color images as BGR by default

img_path = Path("cat.png")

if not img_path.exists():
    raise FileNotFoundError(f"Could not find image at: {img_path.resolve()}")

bgr = cv2.imread(str(img_path), cv2.IMREAD_COLOR)
if bgr is None:
    raise ValueError("cv2.imread returned None. The file may be unreadable or

print("Loaded:", img_path)
print("Shape (H, W, C):", bgr.shape)
print("dtype:", bgr.dtype)
```

```
Loaded: cat.png
Shape (H, W, C): (1076, 1638, 3)
dtype: uint8
```

In [28]:
```
# Convert BGR -> RGB
rgb = cv2.cvtColor(bgr, cv2.COLOR_BGR2RGB)
```

```
# Check here:
# OpenCV stores channels last. This just shows that conversion ran
print("First pixel BGR:", bgr[0, 0].tolist())
print("First pixel RGB:", rgb[0, 0].tolist())
```

```
First pixel BGR: [21, 23, 23]
First pixel RGB: [23, 23, 21]
```

In [29]:

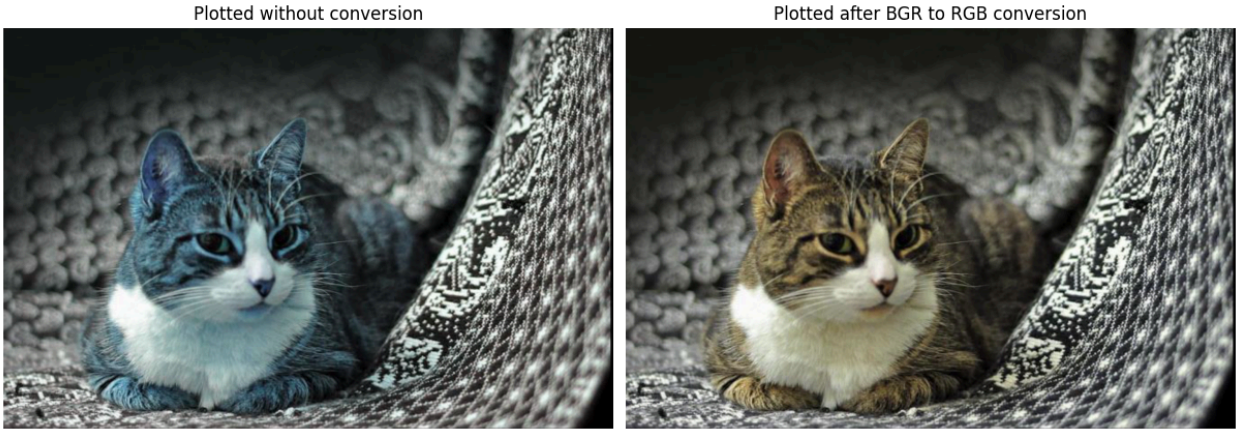
```
# Here we plot using Matplotlib

fig, ax = plt.subplots(1, 2, figsize=(12, 5), constrained_layout=True)

ax[0].imshow(bgr)  # wrong colors (BGR interpreted as RGB)
ax[0].set_title("Plotted without conversion")
ax[0].axis("off")

ax[1].imshow(rgb)  # correct colors (BGR to RGB conversion)
ax[1].set_title("Plotted after BGR to RGB conversion")
ax[1].axis("off")

plt.show()
```



OpenCV uses BGR channel order for color images; Matplotlib expects RGB.

This is why `cv2.cvtColor(bgr, cv2.COLOR_BGR2RGB)` is needed before `plt.imshow(...)`

From here, it would be pretty easy to save the converted image back to disk (e.g., `cat_rgb.png`).

Notes of AI usage

OpenAI's GPT-5.2 model was used to review this code after it was written by a human (me, Adam Nelson-Archer).

The import checking cell was generated by GPT-5.2.